

SupplyChainOS

AI-Powered Supply Chain Resilience & Digital Twin Control Tower

Turning supply chain disruptions from crises into managed events

Team: Incognito Agents	Track: AI	195 tests . 0 failures	35+ API endpoints
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ML Risk Scoring	Digital Twin	Human-in-the-Loop	Explainable AI	Bob Copilot (MCP)	Audit Trail
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Supply Chain Disruptions Are Invisible Until It Is Too Late

SupplyChainOS: See Everything. Decide Confidently. Act Safely.

Disruption Intelligence Real-time geographic + route matching across all active shipments	Digital Twin In-memory what-if simulation; zero impact on operational state
AI Risk Scoring Deterministic rules + RandomForest ML predict delay probability with explainable factors	Business Impact SLA penalty, holding cost, cargo value at risk in real dollars
Cold Chain Monitor Live temperature telemetry with excursion alerts before delivery failure	Recommendation Engine Prioritized, explainable actions with reasoning factors
Fleet Intelligence Idle asset identification + redeployment distance scoring	Human Approval Gate No AI executes high-value decisions autonomously
Route Optimization Haversine-based alternative routes around disruption zones	Immutable Audit Trail Every AI decision and human action recorded
Carrier Recommendation Cost + reliability + capacity scoring across carrier graph	Bob Copilot (MCP) Natural-language queries to structured MCP tool calls to live data

Clean Layered Architecture - No Framework Lock-in

5 Presentation	Next.js 14 Control Tower	App Router, Leaflet, Recharts, TypeScript	COMPLETE
4 Integration	Bob Copilot + MCP Server	Local MCP JSON-RPC, 9 allowed tools, Security layer	COMPLETE
3 API	FastAPI Backend	35+ endpoints, Pydantic v2, Async SQLAlchemy	COMPLETE
2 Engines	11 Pure Python AI Engines	Zero DB deps: DisruptionImpact, ShipmentRisk, PredictiveRisk, ColdChain, Fleet, RouteOpt, Carrier, Cascade, Business, Recommendation, DigitalTwin	COMPLETE
1 Data	PostgreSQL 16	8 tables: shipments, disruptions, fleet, carriers, routes, temp_logs, recommendations, decision_audits	COMPLETE
0 Infrastructure	Docker Compose	3 services: database, backend, frontend	COMPLETE

Key Invariant: Engines have ZERO database dependencies - pure Python math, fully testable in isolation.

Two-Layer Risk Intelligence: Deterministic + ML

DETERMINISTIC ENGINE		ML PREDICTION ENGINE	
Rule-Based Multi-Factor Scoring * Geographic proximity to disruption zone * SLA time remaining vs estimated delay * Cold chain sensitivity + excursion history * Carrier reliability historical score * Route overlap with disruption polygon * Cargo value at risk (SLA penalty estimate) Output: risk_score in [0.0, 1.0]		RandomForest Delay Predictor * scikit-learn RandomForestClassifier * 12 supply chain risk feature inputs * In-process inference - no external API * Feature importance produces reasoning factors * Delay probability output per shipment * Trained on synthetic supply-chain dataset Output: delay_probability + top factors	
Explainable AI: reasoning_factors: [{factor: port_congestion_index, weight: 0.91}, {factor: sla_hours_remaining, weight: 0.78}, {factor: cold_chain_sensitivity, weight: 0.65}]		195 Tests 0 failures	35+ API endpoints

Digital Twin: Simulate Before You Act

DIGITAL TWIN ENGINE	LIVE STATS
Pure In-Memory Simulation Operator configures a what-if scenario: disruption severity, affected routes, carrier changes. Engine runs full risk + impact pipeline on cloned data, returning baseline vs simulated comparison. HYPOTHETICAL label - zero DB mutations at all times Cascade Impact Analysis 2nd-order cascade: disruption affects shipments, fleet resource contention, carrier capacity pressure, financial exposure chain Supporting Optimization Engines Route: Haversine scoring, delay delta, disruption zone avoidance Carrier: Reliability x capacity x cost composite score, ranked alternatives Fleet: Idle asset detection, redeployment distance scoring Impact: SLA penalty dollars, holding costs, cargo value at risk	200 Active shipments 3 Live disruptions 11 AI engines 35+ API endpoints 195 Tests passing

Governed AI: Human Authority, AI Intelligence

HUMAN-IN-THE-LOOP	EXPLAINABLE AI	BOB COPILOT (MCP)
5-State Recommendation Lifecycle PENDING - AWAITING APPROVAL - APPROVED / REJECTED Mandatory human approval gate for high-value decisions. Actor name + timestamp recorded for every action. 4 REST endpoints: /approve, /reject, /override, /execute 6 rule-based approval triggers	Reasoning Factors Per Decision port_congestion: 0.91 sla_exposure: \$42K cold_chain: HIGH carrier_reliability: 0.94 delay_probability: 78% Ordered by contribution weight - not a black box	Natural-language via local MCP User: Which shipments are at risk? - MCP tool: get_high_risk_shipments - FastAPI live DB query - Structured explainable response Cannot approve autonomously Cannot execute SQL 9-tool strict allowlist Input sanitization layer

Immutable Audit Trail - decision_audits table: append-only, read-only UI. Records: AI recommendation generation, human approval/rejection, Copilot queries, actor + timestamp + outcome.

From Crisis to **Managed Event** in Minutes

\$500K+ Cold chain cargo loss prevented per excursion caught early	72h to 8h Delay reduction via AI route recommendation	200 Shipments monitored with zero manual tracking	100% AI decisions governed by human approval + audit
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Demo Highlights * Live disruption - instant cascade impact scoring * Cold chain breach caught 47 min early * Digital Twin: 2x scenario shows \$180K additional exposure * Supervisor approves recommendation - audit trail entry created * Bob Copilot: natural query - MCP - live explainable answer	Why SupplyChainOS Wins COMPLETE: 11 engines, 35+ APIs, full UI VERIFIED: 195 tests passing, 0 failures GOVERNED: Human approval + immutable audit EXPLAINABLE: Every AI decision has reasoning factors SAFE: Digital Twin never mutates operational data
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SupplyChainOS - Turning supply chain disruptions from crises into managed events

Run: cd src and docker compose up --build . Frontend: http://localhost:3000 . API: http://localhost:8000/docs